

BASH PARAMETER EXPANSION

Parameter Expansion Syntax	
Syntax	Description
<code>\${parameter%suffix}</code>	Remove suffix
<code>\${parameter#prefix}</code>	Remove prefix
<code>\${parameter%%suffix}</code>	Remove long suffix
<code>\${parameter##prefix}</code>	Remove long prefixx
<code>\${parameter/pattern/string}</code>	Replace first pattern match with string
<code>\${parameter//pattern/string}</code>	Replace all the pattern with string
<code>\${parameter/%pattern/string}</code>	Replace suffix pattern with string
<code>\${parameter/#pattern/string}</code>	Replace prefix pattern with string

Default Values	
<code>\${FOO:-val}</code>	Uses val if \$FOO is unset, without changing FOO
<code>\${FOO:=val}</code>	Sets \$FOO to val if unset, changing FOO
<code>\${FOO:+val}</code>	Uses val if \$FOO is set (does not change FOO)
<code>\${FOO:?message}</code>	Show message and exit if \$FOO is unset

Substrings	
<code>\${FOO:0:3}</code>	Substring (<i>position</i> , <i>length</i>)
<code>\${FOO:(-3):3}</code>	Substring from the right

Length	
<code>\${#FOO}</code>	Length of \$FOO

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String Substitutions

```
1 food=           # => food is set to an empty
2                 # string
3
4 echo ${food:-Cake} # => Since $food is empty,
5                 # outputs "Cake"
6
7 STR="/path/to/foo.cpp"
8
9 echo ${STR%.cpp}  # => /path/to/foo
10 echo ${STR%.cpp}.o # => /path/to/foo.o
11 echo ${STR%/*}    # => /path/to
12
13 echo ${STR##*.}   # => cpp (extension)
14 echo ${STR##*/}   # => foo.cpp (basepath)
15
16 echo ${STR#*/}    # => path/to/foo.cpp
17 echo ${STR##*/}   # => foo.cpp
18
19 echo ${STR/foo/bar} # => /path/to/bar.cpp
```

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String Slicing

```
1 name="John"
2
3 echo ${name}      # => John
4 echo ${name:0:2}  # => Jo
5 echo ${name::2}   # => Jo
6 echo ${name::-1}  # => Joh
7 echo ${name:(-1)} # => n
8 echo ${name:(-2)} # => hn
9 echo ${name:(-2):2} # => hn
10
11 length=2
12 echo ${name:0:length} # => Jo
```

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Basepath & Dirpath

```
1 SRC="/path/to/foo.cpp"
2
3 BASEPATH=${SRC##*/}
4 echo $BASEPATH      # => "foo.cpp"
5
6 DIRPATH=${SRC%$BASEPATH}
7 echo $DIRPATH       # => "/path/to/"
```

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String Transformation

```
1 STR="HELLO WORLD!"
2
3 echo ${STR,}      # => hELLO WORLD!
4 echo ${STR,,}     # => hello world!
5
6 STR="hello world!"
7
8 echo ${STR^}      # => Hello world!
9 echo ${STR^^}     # => HELLO WORLD!
10
11 ARR=(hello World)
12
13 echo "${ARR[@],}" # => hello world
14 echo "${ARR[@]^}" # => Hello World
```